

EPFO REIMAGINED

Citizen-First Agentic Service Platform

Complete flows, UX rationale, security, crash recovery, risks, and hackathon critique

Core proposition: Instead of making citizens learn EPFO terminology and navigate a complex service tree, the platform starts with the citizen's intended outcome. Users can let an agent orchestrate the workflow or complete it themselves through a redesigned self-service interface.

Design principle: One platform. Two ways to get things done.

1. Executive Summary

The redesign has four pillars: task-first discovery, agentic execution, secure continuity, and user control. It does not remove conventional EPFO interaction; it adds a simpler path while retaining self-service.

Pillar	Meaning	Benefit
Task-first UX	Start with “What do you need?”	Less terminology and navigation burden.
Agent Mode	Intent → dynamic plan → execution → confirmation → Outcome	Guiding-oriented service completion.
Self-Service	User navigates redesigned service categories.	Choice and control.
Secure Continuity	Checkpoints, session ownership, re-authentication, safe session	Reduced work and lower session risk.
Document Vault	Permissioned reusable references.	Less repeated data entry.

Current-state context: the official EPFO member portal exposes UAN login, “Know Your UAN,” application tracking and UAN-related services; it also currently directs UAN activation/allotment through UMANG using Aadhaar-based Face Authentication. [cite]turn0search1[turn0search2]

The official EPFO passbook is a separate surface for EPF balance and claim status, while official member documentation describes functions such as passbook, transfer claims, profile/KYC and related services. [cite]turn0search0[turn0search13]

Scope: this is a product/UX proposal. Real APIs, eligibility rules, authentication methods, data permissions and statutory requirements must be validated before implementation.

2. First-Time User Flow

USER ARRIVES

↓

SELECT LANGUAGE

↓

“What do you need help with?”

↓

Withdraw PF | Transfer PF | Check Balance | Find UAN | Update Details | Not Sure

↓

IDENTITY DISCOVERY / VERIFICATION

↓

Do you have a UAN? → YES / DON'T KNOW

↓ ↓

Verify UAN Help find UAN

↓ ↓

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↓

BASIC PROFILE SETUP

↓

DOCUMENT VAULT

↓

DASHBOARD

↓

AGENT MODE / SELF-SERVICE MODE

The key change from the original sketch is moving the citizen's goal ahead of UAN terminology. UAN can be discovered or verified when required. The official portal already exposes “Know Your UAN” as a distinct function, supporting this mental model. [\[cite turn0search1\]](#)

3. Returning User Flow

```
RETURNING USER
↓
WELCOME BACK
↓
AUTHENTICATE
↓
VERIFY SESSION / DEVICE CONTEXT
↓
LOAD PROFILE + PERMITTED DATA REFERENCES
↓
DASHBOARD
↓
“What would you like to do today?”
↓
AGENT MODE / SELF-SERVICE MODE
↓
IF UNFINISHED TASK EXISTS: Continue / Start Over
```

The dashboard should prioritize active tasks and common outcomes. The current official portal explicitly handles the case where a session is active in another window/browser/system; the redesign should turn session state into a visible, understandable feature. [\[cite turn0search1 turn0search2\]](#)

4. Main Dashboard

Area	Purpose
Primary action	“What would you like to do today?” via text or voice.
Quick services	Balance, claim, transfer, profile/KYC, grievance, tracking.
Active tasks	Incomplete workflows with resume controls.
Document Vault	Permissions, verification state and recent use.
Security	Recent sessions, revoke session, re-authentication.
Self-Service	Open structured service categories.
Help	Official guidance and human escalation.

The dashboard should answer three questions immediately: What can I do? What am I currently doing? What happened to my previous request?

5. Agentic Flow

USER
↓
CHAT / VOICE
↓
INTENT UNDERSTANDING
↓
AGENT PLANNER
↓
"HERE'S WHAT I'LL DO"
↓
DYNAMIC TASK PLAN
↓
EXECUTE → CHECKPOINT → EXECUTE → CHECKPOINT ...
↓
VERIFY RESULT
↓
RESULT / TRACKING

Example: "Check my PF balance"

1. Verify identity / session
2. Retrieve permitted PF account information
3. Fetch latest available balance
4. Present result + timestamp/source context

Example: "I want to withdraw my PF"

1. Verify identity
2. Determine applicable claim path
3. Check eligibility / prerequisites
4. Verify profile/bank information
5. Gather permitted documents
6. Prepare claim
7. Review
8. Explicit confirmation + appropriate authentication
9. Submit / reference
10. Track status

Do not hard-code every task to four steps. The agent plan should be dynamic and dependent on intent, account state, eligibility and missing information.

6. Agent Transparency

State	User-facing behavior
Planned	"I'll verify your identity and check the required conditions."
In progress	"Verifying your bank details..."
Needs user	"Please confirm the destination account."
Sensitive action	"This will submit your claim. Review before continuing."
Completed	"Claim submitted. Reference: XXXX."
Failed	"This step failed. Your saved progress is safe."
Recoverable	"You can continue from Bank Verification."

The agent should expose actions, status and reasons, not hidden internal reasoning. This builds trust without revealing chain-of-thought.

7. Traditional / Self-Service Flow

Use Self-Service or Explore Services in the UI instead of “Legacy Mode.” The latter sounds obsolete.

```
USER
↓
SELECT LANGUAGE
↓
EXPLORE SERVICES
↓
Claims & Withdrawal | PF Transfer | Balance & Passbook | UAN Services
Profile & KYC | Documents | Grievances | Track Applications
↓
SELECT SERVICE
↓
AUTHENTICATE
↓
SERVICE-SPECIFIC WORKFLOW
↓
ELIGIBILITY / VALIDATION
↓
FORM + DOCUMENTS
↓
REVIEW
↓
CONFIRM + APPROPRIATE AUTHENTICATION
↓
SUBMIT
↓
REFERENCE / TRACKING
```

Example: PF withdrawal: Claims → Withdraw PF → Authenticate → Eligibility → Details/documents → Bank/profile verification → Review → Confirm → Submit → Track.

This is still a redesigned traditional experience, not a copy of the old site. Official EPFO documentation and current portal materials show menu/service concepts such as passbook, transfer claims, profile/KYC, service history and online services. [\[cite\]](#)[turn0search13](#)[turn0search15](#)[turn0search17](#)

8. Agent vs Self-Service

Dimension	Agent Mode	Self-Service
Entry	Tell me what you need	Explore services
Navigation	Agent orchestrates	User navigates
Knowledge	Low	Medium
Forms	Agent guides/pre-fills	User completes
Documents	Agent identifies references	User selects
Steps	Dynamic	Structured
Sensitive actions	Confirmation + step-up auth	Confirmation + step-up auth
Recovery	Resume agent task	Resume application
Best for	First-time/complex/language/voice	Experienced/control-oriented users

Both modes should converge on the same secure identity, workflow, data, document and tracking services. The interaction model changes; the security boundary does not.

9. Security & Session Hijacking

Critical correction: a session ID is not proof of identity. Keep session identification, authentication and authorization separate.

```
AUTHENTICATE USER
↓
CREATE SECURE SESSION
↓
RANDOM OPAQUE SESSION ID + SHORT-LIVED CREDENTIALS + SERVER-SIDE STATE
↓
AGENT / SELF-SERVICE TASK
↓
RISK CHECK
↓
LOW RISK → CONTINUE
HIGH RISK → STEP-UP AUTH + EXPLICIT CONFIRMATION
```

Controls: cryptographically random opaque session IDs; secure HttpOnly/Secure cookies where appropriate; expiry; rotation after important authentication events; session revocation; concurrent/suspicious-session handling; server-side authorization; security audit events; risk-based step-up authentication.

EPFO's current official member surface warns users not to share UAN/password and warns against sharing Aadhaar, PAN, bank details or OTPs. The redesign should preserve and strengthen that security posture.

□cite□turn0search2□turn0search0□

10. Crash Recovery / Secure Continuity

TASK STARTS
↓
STEP 1 ✓ → CHECKPOINT
↓
STEP 2 ✓ → CHECKPOINT
↓
STEP 3 → CHECKPOINT
↓
CRASH / NETWORK FAILURE
↓
USER RETURNS
↓
RE-AUTHENTICATE
↓
VERIFY SESSION OWNERSHIP
↓
LOAD LATEST VALID CHECKPOINT
↓
"You were at Step 4 of 6."
↓
CONTINUE / START OVER / DISCARD

Checkpoint field	Example
Task ID	Opaque identifier
Task type	PF withdrawal
Current step	Bank verification
Completed steps	Identity, eligibility, profile
State version	12
Data references	Secure vault references, not raw files
Operation IDs	Unique IDs for side-effecting actions
Last checkpoint	Timestamp + integrity metadata

The session becomes a resumable workflow, not merely a login state. This is one of the strongest product differentiators in the concept.

11. Duplicate Prevention / Idempotency

USER CONFIRMS ACTION
↓
GENERATE UNIQUE OPERATION ID
↓
SERVER PROCESSES REQUEST
↓
SAVE RESULT AGAINST OPERATION ID
↓
RESPONSE LOST / CRASH
↓
USER RETURNS
↓
CHECK OPERATION STATUS
↙ ↘
COMPLETED NOT COMPLETED
↓ ↓
SHOW RESULT RESUME / RETRY SAFELY

This prevents a retry after a network failure from accidentally creating a duplicate side effect. It is particularly important around submissions or other consequential operations.

For the prototype, describe this as an operation ledger + idempotency layer. The real implementation depends on the capabilities of the underlying EPFO integration.

12. Document Vault

DOCUMENT VAULT

- └ Data/document reference
- └ Verification state
- └ Source
- └ Permission / consent scope
- └ Created / updated time
- └ Retention / deletion policy

↓

AGENT REQUESTS REQUIRED DATA

↓

POLICY CHECK

↓

CONSENT WHERE REQUIRED

↓

RETURN MINIMUM NECESSARY DATA

Do not copy raw Aadhaar, bank details or document images into every agent checkpoint. Store protected data separately and use scoped references. This reduces duplication and limits the blast radius of a compromised workflow state.

The user should be able to see what is stored, why it is stored, where it may be reused, and what permissions apply.

13. Shared Architecture

USER
↓
CHAT / VOICE / UI
↓
EXPERIENCE LAYER: AGENT | SELF-SERVICE
↓
IDENTITY + SESSION: AUTH / RISK / AUTHZ
↓
WORKFLOW ENGINE: PLAN / EXECUTE / VALIDATE / TRACK
↓
CHECKPOINT STORE + POLICY ENGINE
↓
SECURE DATA / DOCUMENT VAULT
↓
EPFO SERVICES / APIs
↓
RESULT + AUDIT

The AI should primarily handle interpretation, planning and guidance. Deterministic policy, authorization, validation and transaction execution should sit behind controlled tools and services rather than being delegated blindly to a generative model.

14. Why This Flow Is Better

Current-style challenge	Proposed flow	Benefit
Service terminology first	Outcome first	Lower cognitive load
Many service surfaces	Unified dashboard	Clearer entry point
UAN knowledge assumed	Discover/verify when needed	Less jargon
Complex manual navigation	Agent orchestration	Fewer decisions
AI-only risk	Agent + Self-Service	Choice and trust
Long workflows are fragile	Checkpoints + resume	Less repeated work
Sensitive actions	Risk-based step-up auth	Stronger protection
Repeated data entry	Permissioned vault references	Less repetition
Opaque processing	Visible activity + reasons	More trust
Ambiguous retries	Operation IDs	Duplicate prevention

The strongest argument is not visual polish. It is that a citizen can express an outcome, understand what will happen, complete it with less cognitive overhead, and safely resume when interrupted.

15. Risks & Mitigations

Risk	Level	Mitigation
Session hijacking	Critical	Secure sessions, expiry, rotation, authZ, risk checks, step-up auth.
Wrong AI action	Critical	Deterministic tools, policy checks, validation, confirmation.
Prompt injection	High	Untrusted-content boundaries, tool allowlists, privilege separation.
Data leakage	Critical	Minimum necessary context, encryption, scoped access, redaction, retention control
Duplicate submission	Critical	Operation IDs, idempotency, durable status checks.
Wrong eligibility guidance	High	Authoritative rules/services are the source of truth.
Over-trust in AI	High	Explain actions, show status, confirm, preserve self-service.
Voice ambiguity	Medium	Confirm critical values in UI/text before submission.
Accessibility gaps	High	Regional languages, voice, simple language, assisted/human escalation.
Vault becomes target	Critical	Encryption, access control, consent, audit, minimization, retention.
API/service outage	High	Graceful degradation, save state, safe retry, escalation.

16. Critical Design Critique

Strong decisions

- Task-first onboarding is the biggest UX improvement.
- Agent + Self-Service is stronger than AI-only because it preserves choice.
- Secure Continuity / crash recovery is a meaningful reliability feature and a strong demo moment.
- Dynamic plans make the agentic claim more credible than a fixed four-step wizard.
- A shared backend avoids building two disconnected products.

Must-fix decisions

- Do not treat a session ID as a security mechanism.
- Do not let the LLM directly execute high-risk financial operations.
- Do not claim full EPFO automation without verified APIs/integrations and permissions.
- Do not store sensitive personal data in every checkpoint or prompt.
- Do not make every workflow exactly four steps.
- Use “Self-Service” or “Explore Services” rather than “Legacy Mode” in the user interface.
- Show the agent's user-facing actions and reasons; do not expose hidden chain-of-thought.

The most important architectural insight is: the agent is an interface to a secure workflow system, not the workflow system itself.

17. Recommended Hackathon Demo

1. New user opens EPFO
- ↓
2. Selects language
- ↓
3. Says: “Mujhe apna PF balance check karna hai.”
- ↓
4. Agent identifies intent
- ↓
5. Agent shows dynamic plan
- ↓
6. User authenticates
- ↓
7. Agent executes + checkpoints
- ↓
8. DELIBERATELY INTERRUPT SESSION
- ↓
9. User returns + re-authenticates
- ↓
10. “Welcome back. You were at Step 3.”
- ↓
11. Resume
- ↓
12. Show result
- ↓
13. Show Self-Service as alternative

If possible, demonstrate a higher-risk action with a clear review/confirmation boundary. Use mock data only in the prototype.

18. Success Metrics

Metric	Measure
Task completion	% of users completing intended task.
Time to first action	Landing → correct workflow.
Navigation burden	Screens/decisions for common tasks.
Drop-off	Where users abandon.
Resume success	% of interrupted tasks successfully resumed.
Duplicate operations	Failures caused by retries.
Agent correction	How often users correct intent/plan.
Trust/comprehension	Can users explain what will happen before confirming?
Security incidents	Unauthorized attempts/session anomalies/data leaks.
Accessibility	Completion across language, voice, mobile and assisted flows.

19. Final Product Thesis

We are not redesigning the EPFO website. We are redesigning how a citizen completes an EPFO task.

The platform combines the control of self-service with the convenience of an agent. It starts with the citizen's outcome, uses secure identity and permissioned data, plans a dynamic workflow, checkpoints progress, protects sensitive actions with stronger authorization, and lets the user safely resume after interruption.

ONE EPFO PLATFORM



MVP priority: (1) task-first dashboard, (2) one convincing agentic workflow, (3) secure checkpoint/resume demo, (4) self-service alternative, and (5) explicit confirmation/security layer.

Current-state sources: official EPFO Member Home/Member Interface, EPFO Passbook, and official EPFO documentation. [\[cite\]](#)[turn0search1](#)[turn0search0](#)[turn0search13](#)